

HEAPLINE · REQUIREMENTS

HeapLine requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heapline		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3 results from HPL-CAL-001 on the constructable design (HPL-DDR-002); R2, R5 and R8 at risk and R9 not met, with decisions for Amish in docs/REVIEW.md
0.2	2026-10-03	Amish Chadha	TRL 2; concept status for each requirement; R10 (overload limit) and R11 (holds when let go) added under HPL-DDR-001
0.1	2026-09-30	Amish Chadha	Initial scaffold

HeapLine requirements

Requirements for the site-kept slide-rescue kit. Targets are unchanged from the scaffold; R10 and R11 were added at TRL 2 (HPL-DDR-001). Results are paper estimates from HPL-CAL-001 on the constructable design, not test results. Requirements that are at risk or not met carry a decision for Amish in `docs/REVIEW.md` and the design decisions register.

Safety: HeapLine is used on unstable ground where a second slide can bury the searchers. It is published as an open engineering reference, not certified rescue equipment. The capstan stores energy in its rope; the overload limit (R10) and the holding pawls (R11) are safety requirements.

Table 1. Requirements and their status at TRL 3.

ID	Requirement	Target	Verification (TRL 4 or later)	Result on paper (HPL-CAL-001)	Status
R1	Deploy fast	Kit carried 100 m (330 ft) and first probe line started within 5 minutes of the alarm	Timed drills with the partner cooperative	4.6 min estimated: run to the shed, open, carry the search wave back, post the lookout, lay boards while probes are joined	Met on paper
R2	Probe depth	Probes reach at least 2.5 m (8.2 ft) into loose waste	Probe trials on a waste test heap	Loosened debris: 2.6 m (full usable depth) by one person. Stiffer debris: one person cannot start; two people reach 1.8 m, held back by tip resistance and the probe's buckling load	At risk
R3	Crawl board support	Board carries a 100 kg (220 lb) searcher with sinkage under 50 mm on loose fill	Load tests on representative fill	44 mm on very loose debris with a 1.5 dynamic factor; bridges a 0.8 m void at 4.0 MPa in the battens	Met on paper
R4	Capstan pull	At least 5 kN (1,124 lbf) pull from a hand capstan on a ground anchor	Pull test on CalRig	5 kN with four people at 114 N each; the shear pin releases at about 6.4 kN (5.1 to 7.6 kN over a batch)	Met on paper
R5	Casualty transport	Sheet stretcher moves a 100 kg (220 lb) casualty 30 m (98 ft) over debris with two to four haulers	Drill with manikin	Level debris: 510 N, 127 N each for four haulers but 255 N each for two. On boards: 153 N each for two	Met with four haulers; at risk with two on debris
R6	Stop rule understood	All drill participants state the stop signals correctly after training	Post-drill check	Six stop signals printed on the lid and on pocket cards	Not verifiable at TRL 3
R7	Storage life	All items usable after 12 months stored in the site box	Annual inspection log at the pilot site	Exterior plywood, painted or galvanised steel, UV-stabilised HDPE, acetal, polyester; alkaline cells stored out of the lamps; no lithium cells	Met on paper by material choice
R8	Portable	No single item over 25 kg (55 lb); whole kit carried by four people	Weighing	Heaviest carried item 21.2 kg (capstan frame). Whole carried kit 151 kg, 37.8 kg each for four in one trip; the search wave alone is 21.5 kg each	At risk
R9	Cost	Parts under USD 400 per site kit	Costed bill of materials	USD 1,599 per site kit; the search core alone is USD 526 and the capstan set USD 449	Not met
R10	Overload limit	Rope tension limited so the rope, capstan and anchor are never loaded past the 8 kN design tension	Shear pin release tests on a batch (TRL 4)	6 mm S275 shear pin releases at 6.4 kN nominal, 7.6 kN at the top of the batch spread; rope factor 4.5 at 8 kN	Met on paper

ID	Requirement	Target	Verification (TRL 4 or later)	Result on paper (HPL-CAL-001)	Status
R11	Holds when let go	The drum cannot run back when the bars are released	Release test at working pull (TRL 4)	Two holding pawls on a 24-tooth ring; one pawl takes the design tension at 77 MPa tooth bearing	Met on paper

■ Assumptions

- Pickers and neighbours will organise a search in the first minutes if they have tools and a plan.
- Burial depths in the first minutes of a slide are within reach of hand probes and shovels in at least some cases.
- Sites or cooperatives can keep a box secure and inspected.
- Formal responders will welcome an organised community search rather than stop it.
- A sound anchor (a tree, a parked vehicle or a structural column) can be found on firm ground near most dumpsites; where none exists the capstan is not used.
- Debris resistance to probing has no published data for dumpsite slides; HPL-CAL-001 brackets it with a loosened and a stiffer case.